

Calculus 2

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Limits and Continuity of Multivariate Functions

Problem 1

Determine the necessary and sufficient conditions on the coefficients a , b , and c such that the quadratic form $f(x, y) = ax^2 + bxy + cy^2$ is of the same order as $g(x, y) = x^2 + y^2$ as $(x, y) \rightarrow (0, 0)$.

Hint 1

To prove two functions are of the same order, you must show that the limit of their ratio is bounded between two positive constants m and M as the variables approach the origin.

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Hint 2

Utilize the transformation to polar coordinates: $x = r \cos \theta$ and $y = r \sin \theta$. Observe how the radius r cancels out in the ratio of the two quadratic forms.

Hint 3

Recall that a continuous function on a compact set (like $\theta \in [0, 2\pi]$) attains its maximum and minimum values. Use the condition $b^2 < 4ac$ to ensure the function never reaches zero.

Problem 2

Evaluate the following limit:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^5 + y^5}{x^3 + y^3}$$

Hint 1

Start by factoring the numerator using the sum of fifth powers formula and the denominator using the sum of cubes formula. Look for a common algebraic factor, $(x + y)$, that can be canceled out to simplify the expression.

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Hint 2

After simplifying, convert the remaining fractional expression into polar coordinates by letting $x = r \cos \theta$ and $y = r \sin \theta$.

Problem 3

A set K in $\mathbb{R}^m$ is called a convex set if for any $\mathbf{x}_1, \mathbf{x}_2 \in K$ and any $0 < t < 1$, we have $(1 - t)\mathbf{x}_1 + t\mathbf{x}_2 \in K$. A function $f : K \rightarrow \mathbb{R}$ is called a convex function if for any $\mathbf{x}_1, \mathbf{x}_2 \in K$ and any $0 < t < 1$, we have

$$f((1 - t)\mathbf{x}_1 + t\mathbf{x}_2) \leq (1 - t)f(\mathbf{x}_1) + tf(\mathbf{x}_2).$$

Prove that any convex function on an open convex set K is locally Lipschitz, that is, for any $\mathbf{x}_0 \in K$, there exists a neighborhood U of $\mathbf{x}_0$ and a constant $L > 0$ such that for any $\mathbf{x}, \mathbf{y} \in U$, we have

$$|f(\mathbf{x}) - f(\mathbf{y})| \leq L\|\mathbf{x} - \mathbf{y}\|.$$

Hint 1

Construct a polytope around $\mathbf{x}_0$ within K to find a bound M via vertex combinations.

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Hint 2

For any $\mathbf{x}, \mathbf{y} \in B(\mathbf{x}_0, r)$ with distance d , extend the ray from $\mathbf{y}$ through $\mathbf{x}$ to a point $\mathbf{z}$ at distance r from $\mathbf{x}$. Verify $\mathbf{z} \in B(\mathbf{x}_0, 2r)$.

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Hint 3

Express $\mathbf{x}$ as a convex combination: $\mathbf{x} = \frac{r}{r+d}\mathbf{y} + \frac{d}{r+d}\mathbf{z}$. Apply the convexity condition, subtract $f(\mathbf{y})$, and use bound M to derive the Lipschitz constant $L = \frac{2M}{r}$.

Differential of Multivariable Functions

Problem 1

Let $f(x, y)$ be a differentiable function on $\mathbb{R}^2 \setminus \{(0, 0)\}$ and k be a non-negative integer. Suppose that for any $(x, y) \neq (0, 0)$, the following equation holds:

$$x \frac{\partial f}{\partial x}(x, y) + y \frac{\partial f}{\partial y}(x, y) = k f(x, y)$$

Prove that f is a homogeneous function of degree k . That is, for any $(x, y) \neq (0, 0)$ and any positive real number t , show that:

$$f(tx, ty) = t^k f(x, y)$$

Hint 1

Fix an arbitrary point $(x, y) \neq (0, 0)$. To connect $f(tx, ty)$ and $t^k f(x, y)$, construct an auxiliary function of a single variable $t > 0$:

$$g(t) = t^{-k} f(tx, ty)$$

Your ultimate goal is to prove that $g(t)$ is a constant function independent of t .

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Hint 2

Calculate the derivative $g'(t)$ using the product rule and the multivariate chain rule. Then, evaluate the given condition (the partial differential equation) at the scaled point $X = tx$ and $Y = ty$. Substitute this relationship into your expression for $g'(t)$ to show that $g'(t) = 0$. Finally, evaluate $g(1)$ to finish the proof.

Problem 2

Find the second-order Taylor formula with Peano remainder for the determinant $\det A$ at $A = I$.

Hint 1

Let $A = I + X$, where $X = (x_{ij})$ represents a matrix of small increments. Your goal is to find the Taylor expansion of the multivariate function $f(X) = \det(I + X)$ around the zero matrix $X = O$ up to the second degree.

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Hint 2

Instead of calculating partial derivatives directly, consider the eigenvalues. If $\lambda_1, \lambda_2, \dots, \lambda_n$ are the eigenvalues of X , the eigenvalues of $I + X$ are $1 + \lambda_1, 1 + \lambda_2, \dots, 1 + \lambda_n$. The determinant is simply the product of these eigenvalues:

$$\det(I + X) = \prod_{k=1}^n (1 + \lambda_k)$$

Hint 3

Expand the product $\prod_{k=1}^n (1 + \lambda_k)$ algebraically, keeping only the terms up to degree 2 in λ_k . Finally, connect the symmetric polynomial sums $\sum \lambda_k$ and $\sum_{i < j} \lambda_i \lambda_j$ back to the matrix invariants $\text{tr}(X)$ and $\text{tr}(X^2)$ to assemble your final Taylor polynomial.

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Hint 4

Finally, the answer is:

$$\det(I + A) = 1 + \text{tr}(A) + \frac{1}{2} [\text{tr}(A)^2 - \text{tr}(A^2)] + o(\|A\|^2).$$

Problem 3

Find the second-order Taylor formulas with Peano remainder and with Lagrange remainder for $f(x, y) = \sqrt{1 + x^2 + y^2}$ at the point $(0, 0)$.

Hint 1

Instead of calculating the partial derivatives of $f(x, y)$ directly, construct an auxiliary single-variable function

$g(t) = f(tx, ty) = \sqrt{1 + t^2(x^2 + y^2)}$. This restricts the function to a line segment connecting the origin $(0, 0)$ to the point (x, y) as t varies from 0 to 1.

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Hint 2

Apply the standard single-variable Maclaurin formula to $g(t)$ up to the second degree, with a Lagrange remainder:

$$g(t) = g(0) + g'(0)t + \frac{1}{2!}g''(0)t^2 + \frac{1}{3!}g'''(\theta t)t^3$$

where $0 < \theta < 1$. The value of $g^{(k)}(0)$ is exactly equal to the k -th order polynomial term of $f(x, y)$ at the origin.

Hint 3

Evaluate the expansion at $t = 1$ so that $g(1) = f(x, y)$. You will find that the terms $g(0) + g'(0) + \frac{1}{2}g''(0)$ exactly form the 2nd-order Taylor polynomial of $f(x, y)$. The final term, $\frac{1}{6}g'''(\theta)$, perfectly corresponds to the 2nd-order Lagrange remainder R_2 evaluated at the intermediate point $(\theta x, \theta y)$.

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Hint 4

Finally, the answer is:

$$\sqrt{1 + x^2 + y^2} = 1 + \frac{1}{2}(x^2 + y^2) - \frac{\theta(x^2 + y^2)^2}{2(1 + \theta^2 x^2 + \theta^2 y^2)^{\frac{5}{2}}},$$

Problem 4

Suppose that the second-order Taylor expansion of a twice continuously differentiable function $f(u, v)$ at the point $(0, 0)$ is given by

$$f(u, v) = 1 + 2u - uv + v^2 + o(u^2 + v^2), \quad (u, v) \rightarrow (0, 0).$$

Let $w = f(x + y + z, xyz)$. Find $\frac{\partial^2 w}{\partial x \partial y}(1, 0, -1)$.

Hint 1

Recall the standard form of the second-order Taylor expansion for a function of two variables at $(0, 0)$:

$$f(u, v) = f(0, 0) + f_u u + f_v v + \frac{1}{2} (f_{uu} u^2 + 2f_{uv} uv + f_{vv} v^2) + o(u^2 + v^2)$$

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Hint 2

Before differentiating w , calculate the values of the intermediate variables $u = x + y + z$ and $v = xyz$ at the given point $(x, y, z) = (1, 0, -1)$. You will find that (u, v) maps perfectly to $(0, 0)$, meaning you can directly use the partial derivatives extracted in Hint 1.

Hint 3

Apply the chain rule to find the first partial derivative

$\frac{\partial w}{\partial x} = f_u \frac{\partial u}{\partial x} + f_v \frac{\partial v}{\partial x}$. Then, differentiate this result with respect to y to find $\frac{\partial^2 w}{\partial x \partial y}$.

Implicit Function and Inverse Mapping Theorem

Problem 1

Prove that the equation

$$x^2 + y^2 + z^2 - 2xyz - 1 = 0$$

defines an infinitely differentiable implicit function $z = z(x, y)$ in a neighborhood of $(0, 0)$ that satisfies $z(0, 0) = 1$. Furthermore, find the second-order Taylor polynomial of $z(x, y)$ at the point $(0, 0)$.

Hint 1

To prove the existence of $z(x, y)$, define

$F(x, y, z) = x^2 + y^2 + z^2 - 2xyz - 1$. Check if the conditions are met at $(0, 0, 1)$:

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Hint 2

The second-order Taylor polynomial at $(0, 0)$ is given by:

$$P_2(x, y) = z(0, 0) + z_x x + z_y y + \frac{1}{2}(z_{xx}x^2 + 2z_{xy}xy + z_{yy}y^2)$$

Note the symmetry of x and y in the original equation, which implies $z_x(0, 0) = z_y(0, 0)$ and $z_{xx}(0, 0) = z_{yy}(0, 0)$.

Hint 3

Let $z(x, y) = 1 + ax + by + cx^2 + dxy + ey^2 + \dots$. Substitute this expression into the original equation:

$$x^2 + y^2 + z^2 - 2xyz - 1 = 0$$

Collect the terms for x, y, x^2, xy, y^2 and set their coefficients to zero to solve for a, b, c, d, e .

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Collect the terms for x, y, x^2, xy, y^2 and set their coefficients to zero to solve for a, b, c, d, e .

Hint 4

After balancing the coefficients for x^2, y^2 , and xy , the second-order Taylor polynomial is:

$$z(x, y) = 1 - \frac{1}{2}x^2 + xy - \frac{1}{2}y^2$$